

0:00–0:20

One sentence on the paper or the topic: where it was published and what it does.

One sentence on why it matters to this audience.

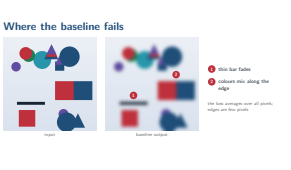
Outline

1	**The problem**	Why does the baseline lose detail?
2	**The method**	How does the method keep edges sharp?
3	**The results**	Does it work?

0:20–0:50

Three sections, each answering one question.

The failure case of section 1 returns at the end of section 3.

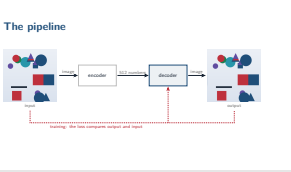


1:00–2:00

Left, an input image; right, the baseline's output for it.

The thin bar fades (marker 1), and the two squares bleed into each other along their shared edge (marker 2).

The baseline is trained with a per-pixel loss, which averages over all pixels; edges are few pixels, so missing them costs little.

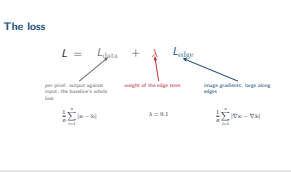


2:10–3:10

The model is an encoder and a decoder: the encoder turns an image into 512 numbers, the decoder turns them back into an image.

Training compares each output with its input, the red dashed path, and updates the model.

The method changes only the loss, the next slide; the decoder is where its effect shows.



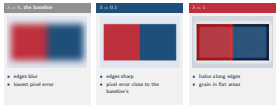
3:10–4:10

The loss adds one term to the baseline's.

The data term compares output and input pixel by pixel; alone, it is the baseline's loss.

The edge term compares their gradients, which are large only along edges, so a blurred edge now costs something.

The weight lambda sets how much; the next slide shows what it changes.



4:10–5:10

The same region, the two squares, for three weights.

Without the edge term the edge blurs; with a weight of 0.1 it stays sharp at almost the same pixel error; with a weight of 1, halos and grain appear.

The weight was chosen on the validation set, not on the test set.

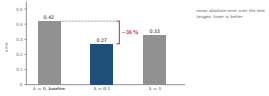


5:10–6:00

Training takes one command and one configuration file.

The configuration names the two loss terms and the weight; everything else is the baseline's.

Evaluation runs on the test set, which training never sees.



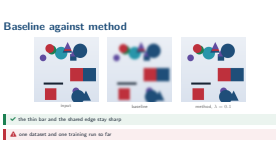
6:10–7:10

The mean absolute error on the test images, lower is better.

With a weight of 0.1 the error falls from 0.42 to 0.27, by about a third.

A weight of 1 is worse than 0.1: the halos of the previous slide cost pixel error.

These are example values; put your own results and their spread here.

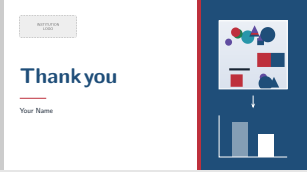


7:10–8:10

The case from the beginning, with the method's output on the right.

The thin bar and the edge between the two squares stay sharp.

The evidence is one dataset and one training run; the spread over several runs is the next check.



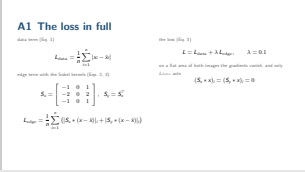
8:10–8:30

Thank you.

Questions to open the discussion: would the edge term help in our own work; how would we check it on several runs; and what would make us trust a sharper output.

Backup slides follow: the loss in full and the references.

A1 The loss in full



If asked

The edge term compares Sobel gradients across and down; on flat areas both are zero, so the term acts only along edges.

The weight 0.1 was the best of five values on the validation set.